Tutorial - 4

Q1 By comparing with general equation of an ellipse i.e. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, determine the semi-major axis, semi-minor axis, the locations of the foci of the ellipse described by $16x^2 + 25y^2 = 400$. Plot the ellipse and mark these parameters on the graph.

Ans: Given, equation

$$16x^2 + 25y^2 = 36 \longrightarrow (1)$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$\Rightarrow \boxed{\frac{x^2}{5^2} + \frac{y^2}{4^2} = 1} \longrightarrow (3)$$

General equation of an ellipse

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \rightarrow (2)$$

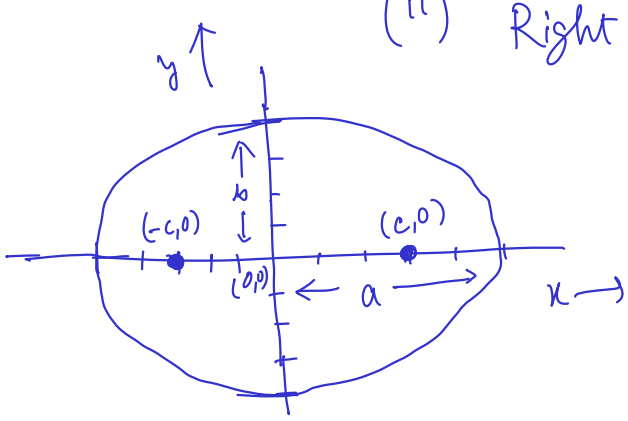
Comparing (2) & (3), we get

Semi-major axis: $a = 5$; Semi-minor axis: $b = 4$

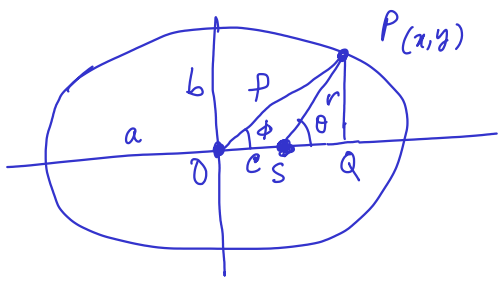
locations of the foci say $(c, 0)$ where $c = \pm\sqrt{a^2 - b^2}$
$= \pm\sqrt{25 - 16}$
$= \pm 3$

(i) Left focus at $(-c, 0) = (-3, 0)$

(ii) Right focus at $(c, 0) = (3, 0)$



Q2 As per Kepler's law, the orbit of a planet around the Sun is an ellipse. Using the equation for an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (in Cartesian coordinate), show that the orbit can be described as $\frac{1}{r}=\frac{1}{\ell}(1+e\cos\theta)$ where the origin of the polar coordinate is at a focus of the ellipse. Express the eccentricity e and semi-latus rectum ℓ in terms of a and b. [Location of focus $(c,0)$ s.t. $c^2=a^2-b^2$]



Eqtn $\boxed{\frac{x^2}{a^2}+\frac{y^2}{b^2}=1}$

location of focus $(c,0)$ s.t. $\boxed{c^2=a^2-b^2}$

$$\boxed{x=p\cos\phi\ ;\ y=p\sin\phi}$$

From diagram: $PQ = p\sin\phi = r\sin\theta$

$OQ = p\cos\phi = SQ + SO = r\cos\theta + c$

$$\Rightarrow \left(\frac{p\cos\phi}{a}\right)^2+\left(\frac{p\sin\phi}{b}\right)^2=1 \Rightarrow \left(\frac{r\cos\theta+c}{a}\right)^2+\left(\frac{r\sin\theta}{b}\right)^2=1$$

$$\Rightarrow b^2(r^2\cos^2\theta+c^2+2rc\cos\theta)+a^2r^2(1-\cos^2\theta)=a^2b^2$$

$$\Rightarrow b^2c^2+2b^2rc\cos\theta+a^2r^2-r^2\cos\theta^2(a^2-b^2)=a^2b^2$$

$$\Rightarrow \underline{b^2a^2}-b^4+2b^2rc\cos\theta+a^2r^2-r^2c^2\cos^2\theta-\underline{a^2b^2}=0 \quad \Big/\ c^2=a^2-b^2$$

$$\Rightarrow (b^2-rc\cos\theta)^2=(ar)^2$$

$$b^2-rc\cos\theta=ar \Rightarrow b^2=ar\left(1+\frac{c}{a}\cos\theta\right)$$

$$\Rightarrow \boxed{\frac{1}{r}=\frac{1}{\ell}(1+e\cos\theta)}$$

with eccentricity: $\boxed{e=\frac{c}{a}=\sqrt{1-\frac{b^2}{a^2}}}$

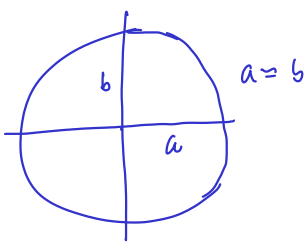
Semi-latus rectum: $\boxed{\ell=\frac{b^2}{a}}$

Q3 Using the derived equation of an ellipse (in polar coordinates), show that for small eccentricity an elliptical orbit can be approximated as a circular orbit. Find the radius of the orbit.

The equation for an ellipse

$$\boxed{\frac{1}{r} = \frac{1}{\ell}(1 + e\cos\theta)}$$

<u>eccentricity</u>: $e = \sqrt{1 - \frac{b^2}{a^2}}$ If $b \to a$ then $e \to 0$



If $e \to 0$ $\quad \frac{1}{r} = \frac{1}{\ell} \Rightarrow \boxed{r = \ell = \frac{b^2}{a} \simeq a}$

The equation of a circle with radius $r = a$